

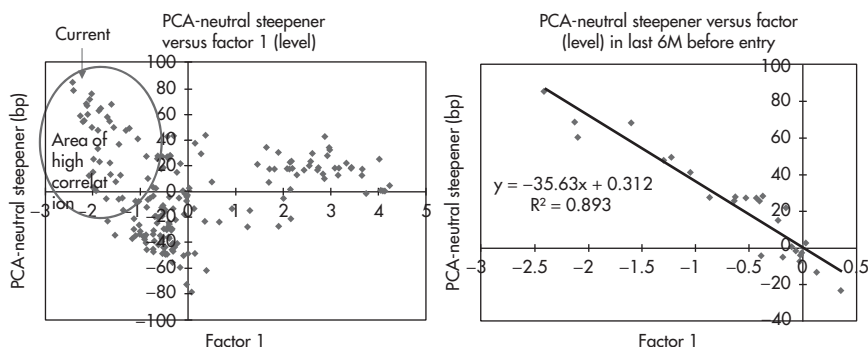


FIGURE 3.23 PCA-neutral steepener versus factor 1.

Sources: data – Bloomberg; chart – Authors.

Data period: 7 Jan 2008 to 22 Aug 2011, weekly data (left chart), 28 Feb 2011 to 22 Aug 2011, weekly data (right chart).

entered on 3 Oct 2011 did not exhibit a significant correlation of factors before entry: the entry level is the isolated point in the lower left corner of Figure 3.14. Unlike the cloud of points along a regression line in Figure 3.23, in the case of the butterfly from Figure 3.14, there is no correlation problem visible – and this was one reason why one trade worked (Figure 3.21) and the other did not (Figure 3.22).

In recent times, the reason for correlation of factors between subperiods is often the shift in the credit assessment of sovereign issuers. Imagine that a PCA is calculated on data before credit risk became an issue in Western government bond markets. From the point in time onwards when it did become an issue, it will impact factor 1 (the yield level tends to increase when credit risk increases) *and* factor 2 (since the longer the maturity, the more it will be affected by increasing credit risk,²⁶ thus an increase in credit risk tends to lead to a higher non-directional steepness of the yield curve). Consequently, the emergence of credit concerns has often resulted in significant correlation between factors 1 and 2. We shall address this issue in detail in Chapter 13 and show how running a PCA on CDS-adjusted yield curves (bond yield minus CDS) can solve the problem.

²⁶As can be seen from Moody's transition matrix, for example, this statement is true for good credits, such as Western government bonds at the start of the recent crisis. When the credit becomes bad, however, the dynamics can change and increasing credit risk can then affect shorter maturities more. See Chapter 13 for more details.

In our experience, ephemeral correlation between factors is the main pitfall of PCA. Conversely, consistently checking for factor correlation, for example, through the method described above, reduces the risk of PCA leading to unsatisfactory trades significantly. When we started using PCA systematically as a tool to find and construct trades in global bond markets about 25 years ago, we did not pay enough attention to the correlation problem and produced a ratio of profitable to overall PCA-based trade ideas of 82%.²⁷ After figuring out that many of the 18% of losing trades were due to the correlation issue, we avoided this trap better and could increase the success ratio to just over 90%, with the number of trades obviously going down. In general, experience is required to strike the right balance between having too many trades and being overly cautious. In the current case, striking this balance requires experience to judge the level of correlation between factors in the subperiod that is acceptable before entering a trade. As a rough guide, Figure 3.14 (isolated current point out of cloud of points with no correlation) and Figure 3.23 (correlated cloud of points leading to current point) provide an illustration of unsuspecting and suspicious situations.

PROBLEMS AND PITFALLS OF PCA 2: INSTABILITY OF EIGENVECTORS OVER TIME

If eigenvectors change after entering into a trade with PCA hedge ratios, the trade will become exposed to unintended factor risk. Imagine again that we have entered into a trade on factor 2 and hedged against factor 1. If the first eigenvector for the time period during which we hold the trade turns out to be different from the first eigenvector calculated on the sample period before the trade (and thus used for determining the hedge ratios for factor 1 neutrality), then we will be hedged against the “wrong” first eigenvector and exposed to the first factor. Thus, a change of the first eigenvector results in the hedge breaking down and in directional exposure. While the cause can be different, the problem is the same as in the case of correlation between factors occurring after entering into a PCA-neutral position: it loses its neutrality and becomes exposed to factors it was not intended to. This section addresses the potential

²⁷We generated these trade ideas by screening mainly Western government bond markets through an approach similar to the step-by-step guide from above, including macroeconomic analysis (as in Table 3.1) when necessary. A trade was counted as profitable when it reverted back to its mean before hitting the stop loss. Since we pursued an analytical goal, we disregarded issues arising in a trading context, such as bid-ask spreads and capital charges.

issue of unstable eigenvectors and offers practical solutions, whose suitability depends on the specific circumstances.

First of all, sometimes a perceived instability of eigenvectors is the result of a too-short sample period for the PCA calculation. This can be easily avoided by choosing a longer time period (step 1) (in case of yield curve analysis usually at least one year).

Moreover, it is important to distinguish changes in factors from changes in factor loadings: changes in factors occur all the time and are no problem for the PCA model and hedges based on it. For example, yields could fluctuate between 1% and 10% (i.e. α_1 exhibits a high volatility), while the eigenvectors remain stable. If the central bank is driving yields both up and down, then the first eigenvector should maintain its downward sloping shape (depicted in Figure 3.6). However, if the central bank cuts rates to zero and then announces that it will maintain a zero policy rate for a number of years, the sensitivity of short rates to directional moves can be expected to decrease versus the sensitivity of long rates. This corresponds to a change in the first factor loading.

Over the last couple of years, many central banks have established a zero-interest-rate-policy (ZIRP) or a similar approach for a significant period of time. This provides an opportunity to study the impact of the shifts between normal central bank policy (with high volatility at the short end) and ZIRP (with low volatility at the short end) on the PCA eigenvectors.

Figures 3.24 and 3.25 depict the evolution of the first eigenvector over time for the US Treasury yield curve, the former using consecutive five-year intervals since 1978, the latter consecutive one-year intervals since 2015. A few observations are worth highlighting:

- In the majority of time periods, the shape of the first eigenvector is downward sloping, i.e. the shorter the rate the higher its sensitivity to changes in the overall yield level. This corresponds to periods with normal central bank policy, when a high degree of uncertainty about the future policy path is reflected in a high volatility of short maturities.
- When the Fed removes the uncertainty about its future policy path, intending to amplify the support of economic recovery, this “unusual” policy is reflected in an “unusual” shape of the first eigenvector, provided that it persists for a sufficiently long period. Correspondingly, the exceptional monetary policy following the financial crisis and lasting for several years has caused exceptionally low levels of rate sensitivity at the short end of the UST yield curve.

Like its cause, i.e. exceptional central bank policy, eigenvector instability is mainly an issue for long-term trades. A curve trade held over several years or even decades would have been affected by the changes depicted in Figures 3.24 and 3.25. As the speed of mean reversion tends to increase with the number of the factor (see above), this problem is less relevant for RV trades on

higher factors. For example, the trade on factor 3 used as example in the step-by-step guide has an average first passage time of 84 calendar days (Figure 3.20), over which time a major Fed policy change is unlikely. In fact, in our experience as RV analysts looking for short-term trading positions, eigenvector instability has seldom had an influence on the performance – unlike the correlation during subperiods described above. By contrast, using PCA for expressing long-term positions on macro-economic cycles via factor 1 does not only suffer from a relatively low expected Sharpe ratio due to the slow speed of mean reversion, but also requires stability of the first eigenvector over the economic cycle. But as the cycle can lead to unusual central bank policy,²⁸ the required stability cannot be taken for granted.

Figure 3.25 suggests that since 2022 the first eigenvector is back to its “normal” shape before ZIRP started. The major exception to this observation is

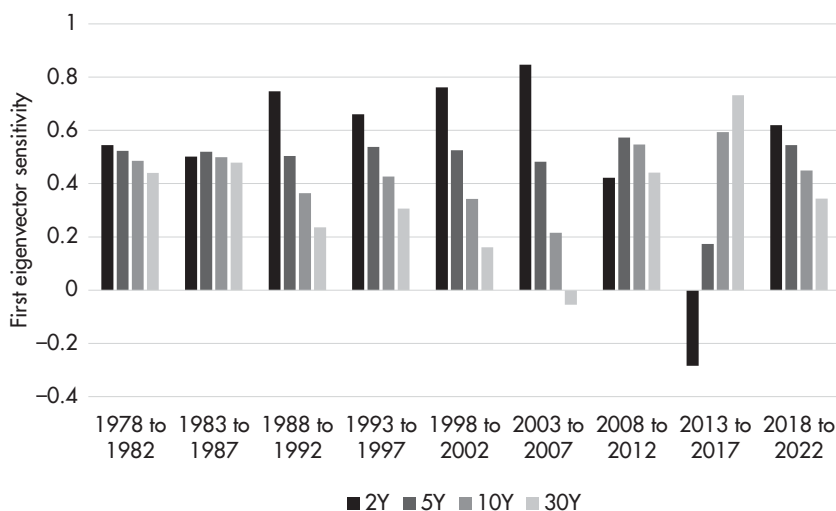


FIGURE 3.24 Evolution of the first eigenvector for US Treasuries since 1978.

Sources: data – Market Yield on U.S. Treasury Securities at Constant Maturity, Quoted on an Investment Basis, retrieved from FRED, Federal Reserve Bank of St. Louis; chart – Authors.

Data period: 1 Jan 1978 to 31 Dec 2022, weekly data, broken down into five-year sections.

²⁸ Recall that policy rates simply going up or down is not a problem; the instability arises when the way in which policy rates increase or decrease changes, for example, when the central bank newly establishes ZIRP.

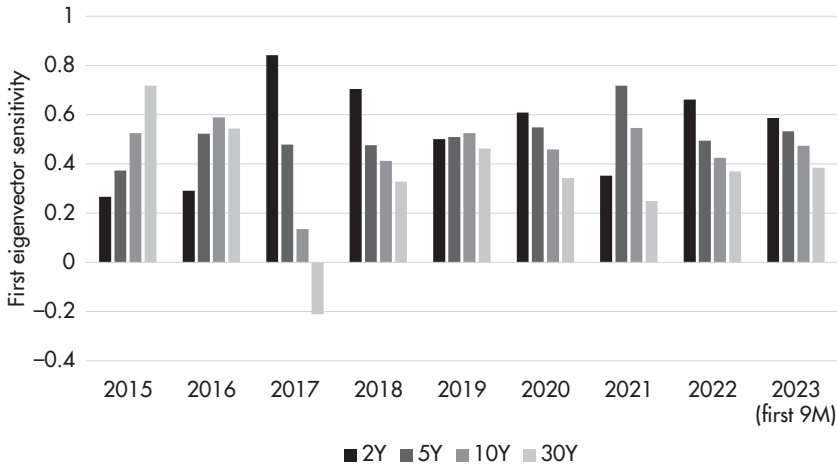


FIGURE 3.25 Evolution of the first eigenvector for US Treasuries since 2015.

Sources: data – Market Yield on U.S. Treasury Securities at Constant Maturity, Quoted on an Investment Basis, retrieved from FRED, Federal Reserve Bank of St. Louis; chart – Authors.

Data period: 1 Jan 2015 to 30 Sep 2023, weekly data, broken down into one-year sections.

Japan, where the first eigenvector remains upward-sloping since the BoJ has implemented ZIRP.²⁹

As a practical matter, for the US it seems advisable to use data from 2022 onwards for calculating PCA, thereby excluding data from the “unusual” policy period while still having a sufficiently large range of observations.

If instability of eigenvectors is a concern, particularly for trades with a relatively long expected holding horizon, an analyst has several possibilities to react:

- As a start, it may be generally a good idea to test the impact of different shapes of the eigenvectors (e.g. from ZIRP) on the performance of the trade. If the performance is sufficient in all reasonable scenarios, the concern about stability can be alleviated.

²⁹Also the introduction and most recent relaxation of “yield curve control,” that is, a cap on 10Y JGB rates, has been visible in the shape of the eigenvectors. An isolated discontinuation of yield curve control could be expected to cause the first eigenvector to become steeper, while discussion about an end to the ZIRP would move Japan closer to the “typical” shape already visible in the other major markets.

- One can try and intentionally position for the short end of the eigenvector to change: we know that a change in the eigenvector will result in the PCA hedge breaking down. But if we can forecast how the eigenvector is going to change, we can position ourselves in such a way that we will end up being overhedged in a falling and underhedged in a rising market (i.e. turn the problem of changing eigenvectors into a profitable trading strategy that has similarities with delta hedging a long option position). For example, if the BoJ ends the zero interest rate policy, an increase in rates (factor 1) should be linked to the short end becoming more volatile (sensitivities in first eigenvector increasing at the short relative to the long end). A curve-flattening position dynamically hedged against (the changing) factor 1 should therefore in fact be overhedged in declining and underhedged in increasing markets.³⁰
- Another method to mitigate the problems of eigenvector instability is to use other input variables, which are less influenced by shifts in monetary policy. For example, the shadow rate may well keep the “usual” volatility characteristics even if central banks switch to “unusual” ZIRP. Also using bond yield minus CDS could be an alternative. We’ll revisit this topic in the relevant Chapters 6 and 13.
- And, of course, if the risk of eigenvector instability is deemed too high for a particular trade to be mitigated by any of these methods, one can simply do without the benefits of PCA.

PCA as a Tool to Construct New Types of Trades

Finally, we provide an example of how PCA could be used in the currency market, thereby underlining its universal applicability. At the same time, we shall illustrate how creatively a PCA can be applied, using the step-by-step guide from above as stepping stones to a trade idea, which only PCA enables but in which PCA is just one of several parts.

When running a PCA on JPY, GBP, SEK, CHF, AUD, and SGD (versus USD), it is useful to adjust the series for the difference in absolute values in order for the charts to be legible (otherwise, e.g. 100 JPY per USD versus 0.6 GBP per USD would show up as huge difference in sensitivities to the eigenvectors). Thus, we have run the PCA on synthetic currencies, starting all at a value of 1 on 4 Jan 1999. The PCA then uses weekly data of these synthetic series from 4 Jan 1999 until 25 July 2011 and can be seen on the Excel sheet on the website accompanying this book.

³⁰This strategy was first published in the ABN AMRO Research note from 17 Nov 2006 “Exploiting the regime shift with PCA weighted flatteners” and is mentioned here with kind permission from RBS.

Figure 3.26 depicts the scaled eigenvalues. It turns out that the relative explanatory strength of the first and second factor in the FX market is similar to the bond market, while the eigenvalues above 3 decrease slower than in the case of bonds.

In Figure 3.27, the eigenvectors are displayed. As in the case of a PCA on bonds, the first eigenvector has only positive entries (i.e. a change in factor 1 affects all currencies similarly). However, the sensitivities may seem puzzling: if factor 1 increases, SEK, CHF, and AUD increase a lot, while JPY, GBP, and SGD less so. Since few FX traders think of grouping currencies in such a manner, this puzzle may represent an interesting, PCA-induced insight, but requires further examination, which will be provided below by looking simultaneously at the time series of factor 1.

Likewise, factors 2 and 3 cause differentiation among currencies (e.g. if factor 2 rises, JPY, CHF, and SGD increase, while GBP, SEK and AUD decrease). This grouping may sound more familiar, as it puts “low risk” and “high risk” currencies together.

Since the interpretation of eigenvectors (i.e. market mechanisms) is less straightforward than in the case of Bunds, we need to consider additionally the time series of the factors and their link to external variables in order to be able to complete the interpretation of the PCA results. The evolution of the factors is shown in Figure 3.28.

This graphical representation may well evoke the visual memory of the analyst. For example, he may find that the evolution of factor 1 is almost a

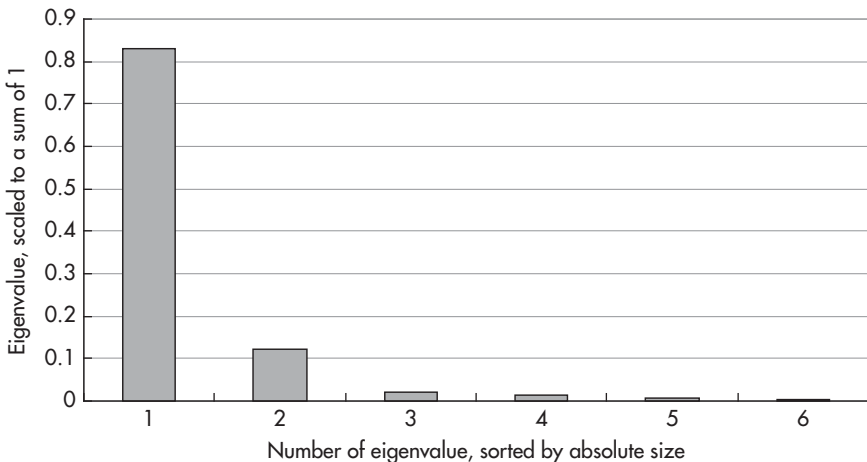


FIGURE 3.26 Scaled eigenvalues of a PCA on currencies.

Sources: data – ECB; chart – Authors.

Data period: 4 Jan 1999 to 25 Jul 2011, weekly data.

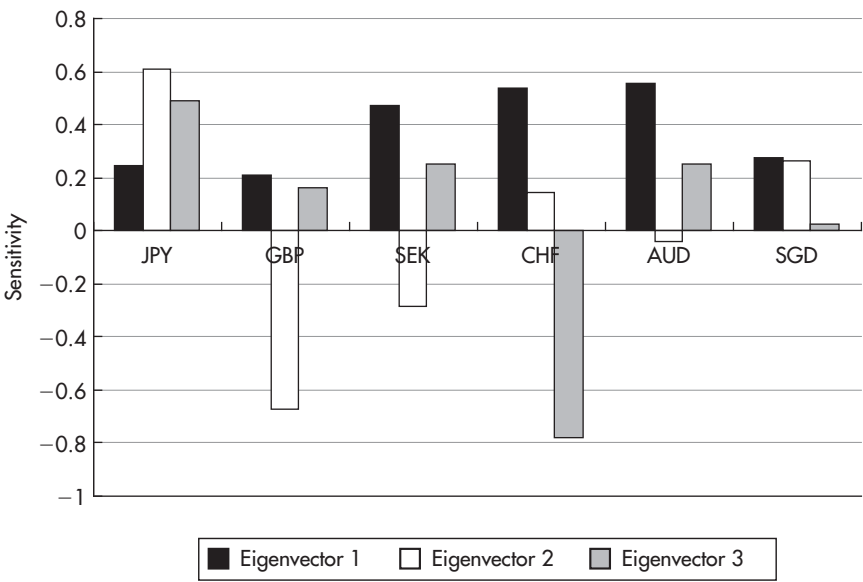


FIGURE 3.27 Eigenvectors of a PCA on currencies.

Sources: data – ECB; chart – Authors.

Data period: 4 Jan 1999 to 25 Jul 2011, weekly data.

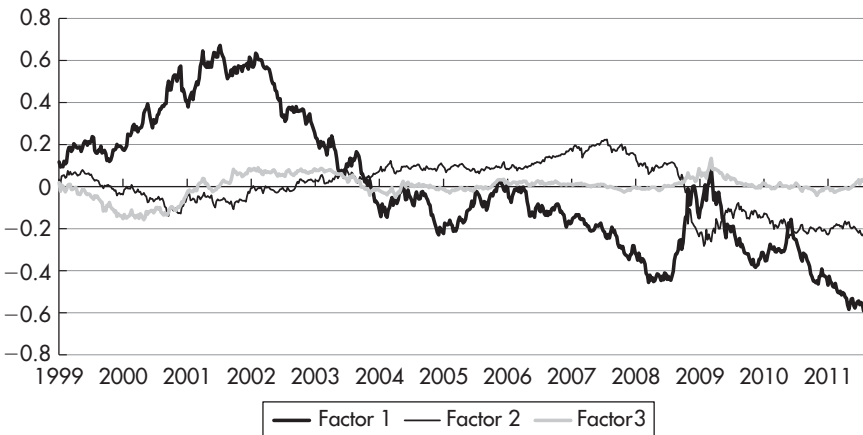


FIGURE 3.28 Factors of a PCA on currencies.

Sources: data – ECB; chart – Authors.

Data period: 4 Jan 1999 to 25 Jul 2011, weekly data.

mirror image of the USD–EUR FX rate. This visual discovery can be confirmed (e.g. via a regression, which returns a correlation of -0.94 between factor 1 and the EUR exchange rate). Thus, an increase of factor 1 is strongly linked to the EUR weakening versus the USD. An interesting consequence of that result is that the exchange rate of the USD versus the EUR (which we excluded from the PCA input series) explains 83% (scaled first eigenvalue) of the moves of the USD versus the other currencies. Hence, one may conclude that the USD–EUR rate is by far the most important driving force of all USD FX rates. Moreover, the sensitivities versus the first factor (first eigenvector) group the currencies with a strong (SEK, CHF, AUD) and not so strong (JPY, GBP, SGD) tendency to weaken versus the USD in case of factor 1 going up (i.e. the EUR weakening against the USD).

The time series of factor 2 corresponds well to a more psychological variable such as “risk on/risk off.” And as expected, factor 2 falling significantly in 2009 (and staying low) has resulted in those currencies with a positive sensitivity to factor 2 (“safe havens”) (JPY, CHF, SGD) to outperform those with a negative sensitivity to factor 2 (“risky”) (GBP, SEK, AUD). This is probably the picture most traders have in mind when they think about the FX market. Note, however, that a PCA relegates this market mechanism to number two, revealing that it only explains 12% of the FX market action – and reveals the more important structure given by eigenvector 1.

In cases of puzzling eigenvectors, it is often useful to calculate a table of heuristic regressions, with the factors as dependent variables and with various candidates serving as independent or explanatory variables. We have tried a couple of those regressions and present the results in Table 3.2.³¹

Table 3.2 confirms our optical interpretation of factor 1 being closely linked to the EUR and factor 2 being a “risk on/risk off” factor, which has therefore some correlation to the direction of USD interest rates (as represented in the first factor of a PCA on USD swaps) and the VIX. Furthermore,

TABLE 3.2 Correlations of the First Three Factors of a PCA on Currencies versus Candidates for External Explaining Variables

	Factor 1	Factor 2	Factor 3
EUR	-0.94	-0.01	0.13
S&P500	-0.20	0.27	-0.66
VIX	0.15	-0.46	0.28
US swap PCA factor 1	0.64	0.55	-0.49
US swap PCA factor 2	0.42	-0.10	0.16
US swap PCA factor 3	-0.13	-0.04	-0.41
Oil	-0.86	-0.12	-0.07

³¹For some series, the time period for the regression is slightly different.

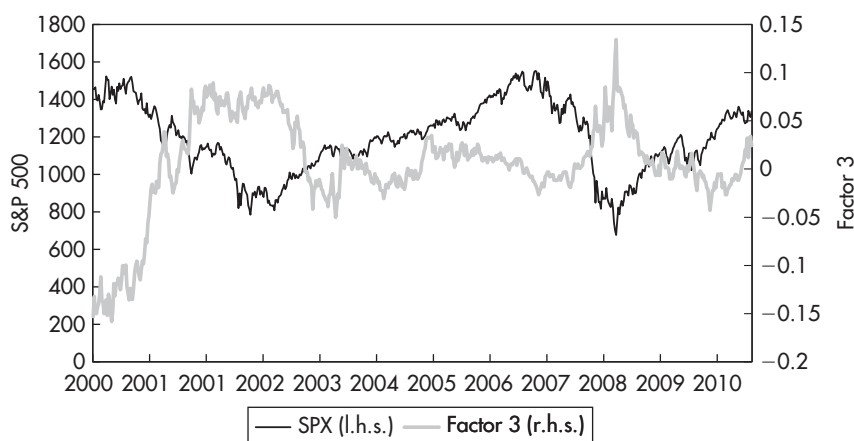


FIGURE 3.29 Factor 3 of a PCA on currencies versus the S&P500 index.

Sources: data – ECB, Bloomberg; chart – Authors.

Data period: 3 Jan 2000 to 25 Jul 2011, weekly data.

Table 3.2 reveals that the third factor is linked to the S&P500 index – and, indeed, taking another look at the time series of factor 3 in the chart above confirms that it is a close mirror image to the stock market. This relationship is depicted in Figure 3.29 in more detail.

These new insights, which a PCA provides into the FX market, could be used to model currencies by a three-factor model, with the factors linked to external variables like EUR, risk adversity (e.g. VIX), and stock prices.

In the following, however, we would like to show how the insights of a PCA into market mechanisms can be used to construct new trading positions, which would not be possible or understandable outside of the PCA framework.

While there is a reasonable correlation between factor 3 and the S&P500, currently (25 July 2011), the residual of a regression is rather high. Figure 3.30 illustrates that the current point is quite far away from the regression line. Note that if we use shorter time horizons for the regression, we still witness a significant residual. This means that the relationship between factor 3 and stock prices, which has been relatively stable over the past 12 years, is currently disturbed. Thus, if we believe that the long-term relationship will hold in future (and we see no reason not to do so), we may want to bet on the residual disappearing.³² Hence, we investigate for a trade of factor 3 of a PCA on currencies versus the S&P500 index.

Since we cannot trade factor 3 directly, we need to find a portfolio of three currencies, hedged against factors 1 and 2 and highly correlated to factor 3. Given the relatively high strength of factor 4 in particular, not every two-factor

³²Ideally, we would like to see an even higher correlation, though.